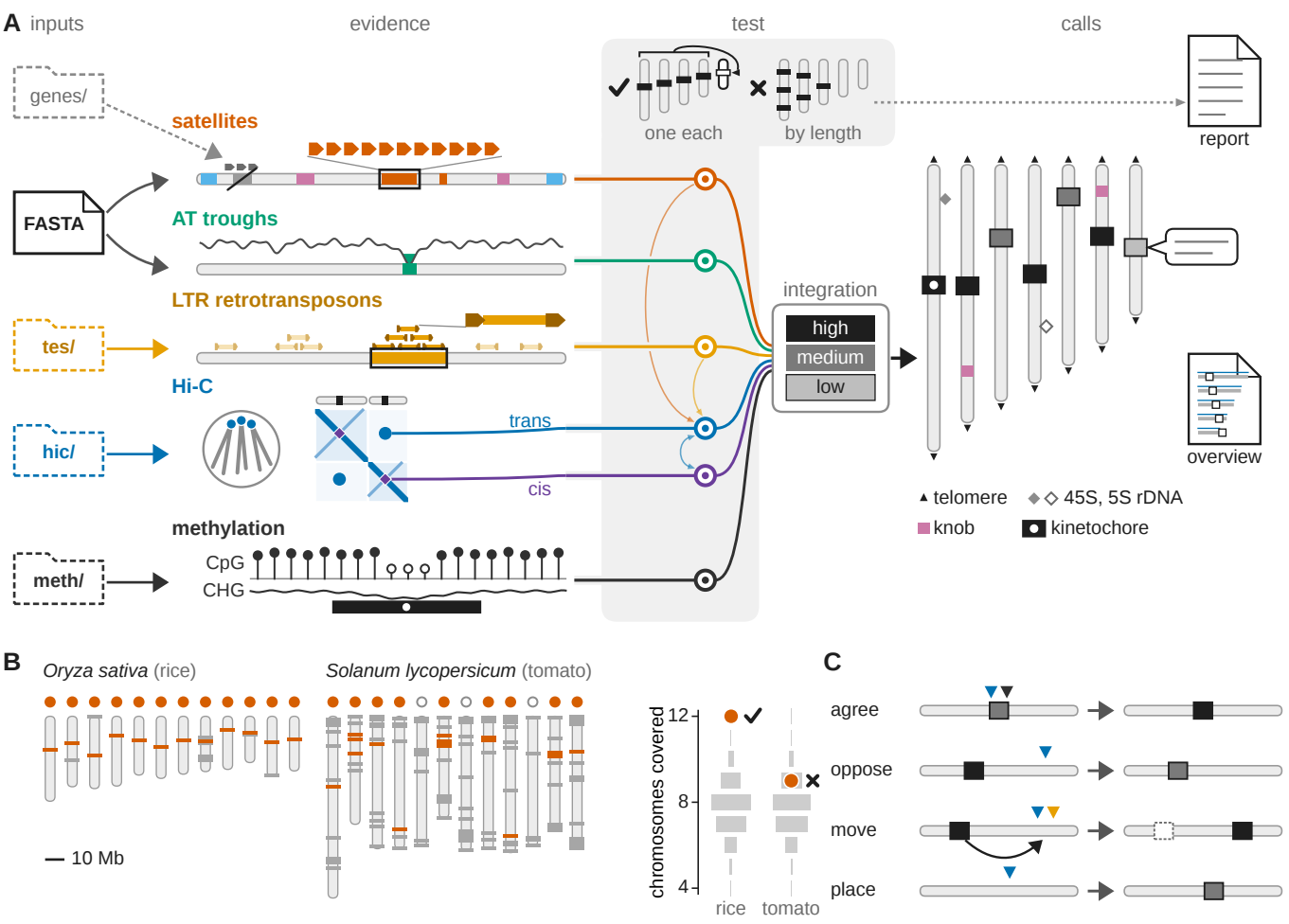

A
inputs
evidence
test
calls
genes/
FASTA
satellites
AT troughs
LTR retrotransposons
tes/
Hi-C
hic/
trans
cis
methylation
meth/
CpG
CHG
one each
by length
report
integration
high
medium
low
overview
telomere
45S, 5S rDNA
knob
kinetochore
B
Oryza sativa (rice)
Solanum lycopersicum (tomato)
10 Mb
chromosomes covered
12
8
4
rice
tomato
C
agree
oppose
move
place

A centromere occurs once per chromosome, so whatever marks centromeres must fall once on every chromosome. cenfuse lets each kind of evidence act only after it passes this test in the genome at hand, and grades every call.

A Five lines of evidence, from the assembly alone or from optional data, each find centromere-like loci on their own: satellite arrays, AT-rich troughs, retrotransposon clusters, Hi-C contacts and DNA methylation. Each must then show, in this genome, that it marks one place per chromosome rather than places scattered by length. Lines that pass are combined into one call per chromosome, graded high, medium or low.

B The test on real satellites. In rice the largest arrays of the centromeric satellite sit once on every chromosome, far beyond chance; in tomato they pile onto some chromosomes and miss others, as chance would, so tomato's satellite calls stay low.

C Tested lines adjust the sequence call: agreement raises it, disagreement lowers it, strong evidence elsewhere moves it (a high call only with two lines), and tested evidence can place a call where the sequence gives none.

Terms. A *line* is one kind of evidence. The *sequence call* is the call from the FASTA: the main centromeric satellite array, or an LTR retrotransposon core when tes/ is given. The *test* asks whether a line marks the other chromosomes' centromeres once each, more often than chance, where chance places marks in proportion to chromosome length. A line that passes may change calls; one that fails is only reported. Confidence: high, medium or low (gray ramp).

A Inputs: the FASTA is required (solid outline); tes/, hic/, meth/ and genes/ are optional (dashed). *Satellites:* tandem-repeat families found from scratch get roles by where their arrays fall (centromeric orange, subtelomeric blue, knob pink); the call is the main centromeric array (black box); families in gene-rich regions are vetoed. *AT troughs:* the deepest GC trough of each chromosome. *LTR retrotransposons* (each with its two LTRs): families concentrated once per chromosome are learned from the other chromosomes ($P \le 0.001$); the densest cluster is the core, and of two comparable clusters the significantly younger one (saturated) wins. *Hi-C:* centromeres cluster in the nucleus and meet in trans (blue spots); the folded arms (anti-diagonal) and the contact across the centromere (cis, purple) are further readouts. *Methylation:* a CpG dip marks the kinetochore inside the call (primates; it changes no call); CHG is a second line in plants. Gray band: the test (top), at $P \le 0.05$ for satellites and AT troughs and $P \le 0.01$ for Hi-C; each chromosome is judged by the others. Thin links: Hi-C is tested against the sequence calls and the LTR cores, its cis readout against its trans one.

B Every cluster of the centromeric satellite families (arrays within 500 kb merged) on each chromosome, both genomes at one scale; clusters are drawn at least 1.7 Mb tall so that all can be seen. Orange: the 12 largest by satellite length; filled dot: a chromosome holding one of them. *Oryza sativa* (T2T Nipponbare): 12 of 12 chromosomes covered (chance median 8; $P = 0.0001$, the smallest possible with 10,000 draws), pass. *Solanum lycopersicum* (Heinz 1706): 9 of 12 (chance median 8; $P = 0.24$), fail, so its satellite calls stay low unless an LTR core agrees. Right: chromosomes covered by chance (gray; bar width = share of 10,000 draws of 12 marks placed in proportion to length) and observed (orange). cenfuse 0.15.0, FASTA only. Tomato's centromeres lie beside its abundant satellite (113 clusters, against rice's 20), not on it: from the FASTA alone 1 of 12 calls is right, all of them low; potato, also with 12 chromosomes, fails the test too.

C Each row: the call before (left) and after (right); triangles mark where a tested line points, in its color. Agree: each agreeing line raises the call one level, two at most. Oppose: a high call that one tested line places elsewhere (more than 4 Hi-C bins away) stays, one level lower. Move: a strong tested peak moves a low or medium call; a high call moves only with a second line at the new place (dashed: the sequence call, still reported). Place: where the sequence gives no call, tested evidence places one (Hi-C or an LTR core: medium; an AT trough or untested Hi-C: low). Over 47 genomes with truth, high calls were right 406 of 410 times, medium 64 of 69, low 118 of 145. Exceptions: the test cannot pass with about 4 or fewer chromosomes; inputs from a plant other than the assembled one never decide; holocentric genomes get no confident call (0 in 196 chromosome-runs).

Analysis: asc-oulab:/data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/v10.3/v10.4/v11/v12/v13/v14 · Script: /data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/v10.3/v10.4/v11/v12/v13/v14/figures/workflow/workflow.py · 2026-11-10